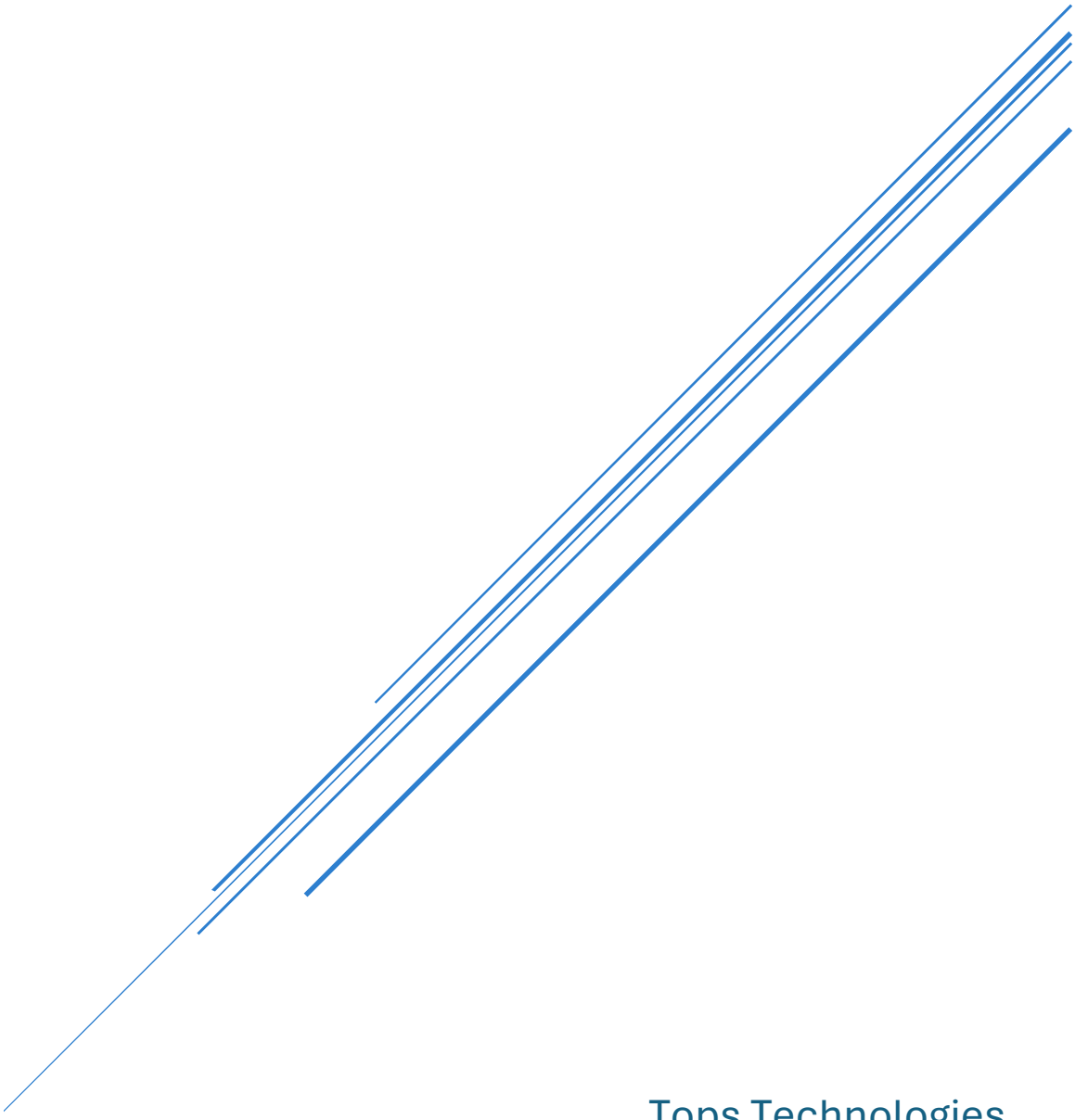
DAY - 1 INTRODUCTION TO NETWORKING AND OSI LAYER

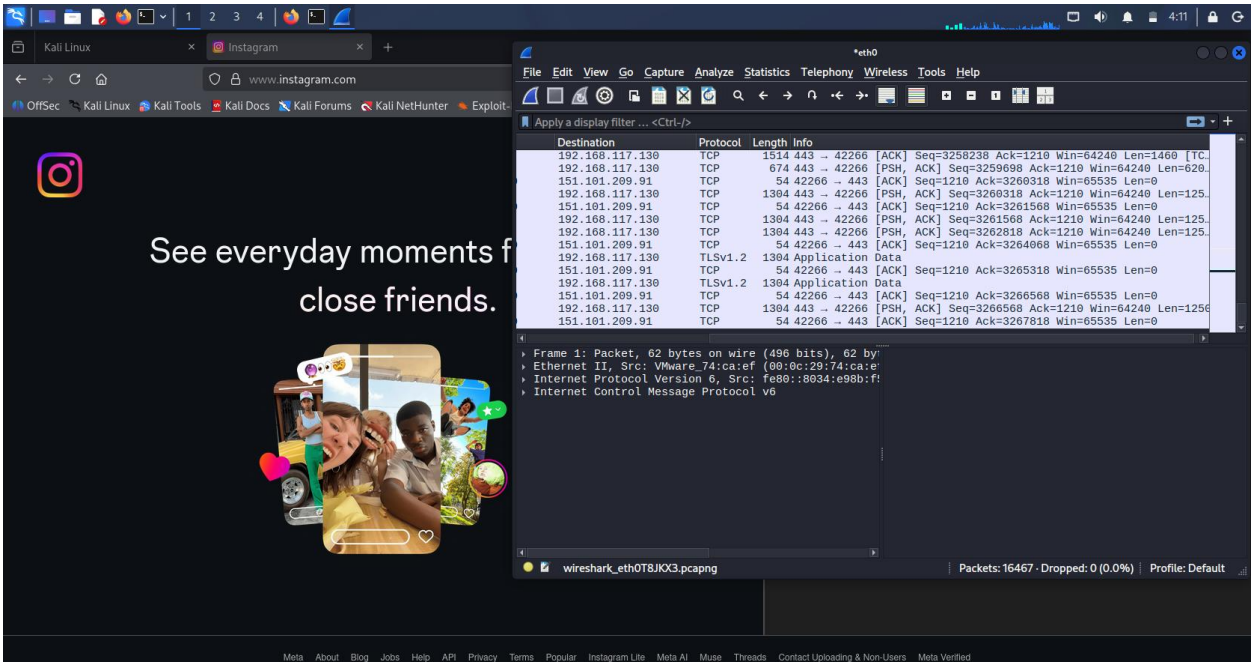
Assignment - 12



Tops Technologies
Cyber Security

1. **Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.**

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While visiting Instagram on Kali Linux, I observed the following protocols in Wireshark:

1. **TCP (Transmission Control Protocol) – Provides reliable data transmission between devices.**
2. **TLSv1.2 (Transport Layer Security) – Encrypts data exchanged with secure websites.**
3. **Ethernet II – Defines how data is transmitted over an Ethernet network.**
4. **IPv6 (Internet Protocol Version 6) – Handles packet addressing and routing.**
5. **ICMPv6 (Internet Control Message Protocol Version 6) – Supports network diagnostics and error reporting.**

2. **Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.).**
   Hint: Use the Protocol column and Info details in Wireshark to help you map protocols to OSI layers.

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Identify the OSI Layers of Three Packets

Packet / Protocol	OSI Layer	Description
1. TCP	Layer 4 – Transport	Provides reliable data transmission between devices.
2. TLSv1.2	Layer 6 – Presentation	Provides encryption and decryption of transmitted data.
3. ICMP	Layer 3 – Network	Reports network errors, such as destination or port unreachable messages.

3. **Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).**

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OSI 7-Layer Model

Layer & Name	Main Function	Real-World Example
7 **Application Layer** (Provides network services to end users)	Handles user applications and provides the interface for network communication.	**HTTP / HTTPS** (Web browsing)
6 **Presentation Layer** (Formats and translates data)	Ensures data is in a readable format, handles encryption and compression.	**SSL / TLS** (Encryption)
5 **Session Layer** (Manages communication sessions)	Establishes, manages and terminates sessions between applications.	**NetBIOS** (Session management)
4 **Transport Layer** (Ensures reliable data transfer)	Provides end-to-end communication, error recovery, and flow control.	**TCP** (Reliable delivery)
3 **Network Layer** (Handles routing and addressing)	Determines the best path for data and handles logical addressing (IP addresses).	**Router** (Forwards packets)
2 **Data Link Layer** (Handles framing and error detection)	Provides node-to-node communication, handles MAC addressing and error check.	**Switch** (Uses MAC addresses)
1 **Physical Layer** (Transmits raw data bits)	Defines the physical medium, signals, and data transmission rates.	**Ethernet Cable** (Wired transmission)

4. **Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words.**
   Hint: Think about how your message is typed, packaged, sent over the internet, and received.

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How a WhatsApp Message Travels Through the OSI Model

When I send a message on WhatsApp, it passes through different layers of the OSI Model before reaching the receiver.

1. Application Layer (Layer 7): I type a message, such as "Hello," in WhatsApp and press Send.

2. Presentation Layer (Layer 6): The message is encrypted so that it is protected during transmission.

3. Transport Layer (Layer 4): Transport protocols help deliver the data between devices. TCP may be used by some services, while WhatsApp can also use other transport methods.

4. Network Layer (Layer 3): IP addressing helps route data packets across networks toward the receiver's device.

5. Data Link Layer (Layer 2): Wi-Fi or Ethernet carries data across the local network using frames.

6. Physical Layer (Layer 1): The data travels as radio signals over Wi-Fi or as electrical or optical signals through cables.

When the message reaches the receiver's device, the data is processed through the layers in reverse order, and the message appears in WhatsApp.